those specialised SOCs which are designed for being used in NAS devices. For example, the Atom CE5315 is one such processor.

RAM

RAM requirements scale according to the features that are utilised. Which means if you have error checking and correction or anything that resembles similarly then the RAM requirements ramp up. There are certain algorithms that need 1GB RAM per TB of data being managed.

Interface

Wired/Wireless

If you wish to stream directly from the NAS device then you could go in for a model having inbuilt WiFi, however, it would be better if you stick to a modular approach and buy a device without the WiFi component. This way you can simply change your wireless router when you wish upgrade to a better 802.11 protocol revision.

Ports

Ports for connecting to various storage peripherals are another aspect to be considered if you are to hook up your external devices to your NAS. So eSATA, USB 3.0 and Thunderbolt are the more popular high-speed ports commonly featured on NAS devices.

Gigabit/100Megabit

The days of the 10/100 interface are behind us and Gigabit is the way to go. Even routers now support Gigabit transfer speeds over 802.11ac which is the latest WiFi protocol. Also having a greater bandwidth means that more clients can be satisfactorily catered to simultaneously.

OS

The operating system of the NAS plays a huge part in the decision process. Most of the features are actually enabled by the software rather than the hardware which is why an OS which is more open source friendly will have that many plugins available to customise your user experience.

For example, you can make a web server or a media server out of your NAS if you manage to get the right plugin. Also, features like remote access over the internet and auto-backup are enabled by the software in most devices. Hot swapping is another feature which is helpful if you plan on swapping the hard drive out of your NAS device on a regular basis.

The flexibility of a NAS in terms of features is heavily dependent on the OS

RAID Level Comparison

Features	RAID 0	RAID 1	RAID 1E	RAID 5	RAID 5EE	RAID 6	RAID 10
Minimum Drives	2	2	3	3	4	4	4
Data Protection	None	Single-drive failure	Single-drive failure	Single-drive failure	Single-drive failure	Two-drive failure	One disk failure per sub-array
Read Performance	High	High	High	High	High	High	High
Write Performance	High	Medium	Medium	Low	Low	Low	Medium
Read Performance (Degraded)	N/A	Medium	High	Low	Low	Low	High
Write Performance (Degraded)	N/A	High	High	Low	Low	Low	High
Capacity Utilisation	100%	50%	50%	67-94%	50-88%	50-88%	50%

Features

RAID

Hardware RAID is preferred over a software RAID anytime. There are multiple RAID configurations for different usage scenarios. Eg. RAID 0 for speed and RAID 1 for redundancy. Depending on what you wish to set up you'll need to pick a NAS with the right number of bays since there is a minimum number of hard drives needed for each RAID configuration.

Printer

If your router doesn't support connecting a printer via USB then you can opt for a NAS device which does support printers via USB. An ethernet port on a printer is a premium feature so it's an uncommon feature in entry level printers.

Noise

The noise factor depends upon the build quality of the NAS since a device with more drives will obviously vibrate a lot more. Excessive vibration is bad for any and all hardware device. This is another reason to go for NAS optimised drives since they intelligently balance the load and thus vibration is greatly curtailed. Nonetheless, you should check that the drive bays are secure and have padding or springs to absorb any vibrations.

Energy Efficiency

Being energy efficient is a feature that is most welcome for any electronic device. This is not only economical in the long run but an energy efficient device also remains cooler. Most NAS drives feature some sort of power management module hard coded into the drive and reduces power during overall usage. d